

Question:

let $\mathbf{M}$ be a 3×3 symmetric matrix with eigenvalues 1,2 and 3. let $\mathbf{N}$ be any 3×3 matrix with real eigenvalues such that $\mathbf{MN} + \mathbf{NM} = 3\mathbf{I}$, where $\mathbf{I}$ is a 3×3 identity matrix. Then which of the following cannot be eigenvalue(s) of the matrix $\mathbf{N}$? (ST 2022)

a) $\frac{1}{4}$

b) $\frac{3}{4}$

c) $\frac{1}{2}$

d) $\frac{7}{4}$

Solution:

let $\lambda_1, \lambda_2, \lambda_3$ be the eigenvalues of $\mathbf{N}$
By Eigenvalue decomposition

$$\mathbf{M} = \mathbf{Q}\mathbf{D}\mathbf{Q}^T \quad (1)$$

where

$$\mathbf{Q}^T \mathbf{Q} = \mathbf{Q}\mathbf{Q}^T = \mathbf{I} \quad (2)$$

$$\mathbf{D} = \text{diag}(1, 2, 3) \quad (3)$$

$$\Rightarrow \mathbf{D} = \mathbf{Q}^T \mathbf{M} \mathbf{Q} \quad (4)$$

$$\mathbf{MN} + \mathbf{NM} = 3\mathbf{I} \quad (5)$$

$$\mathbf{Q}^T (\mathbf{MN} + \mathbf{NM}) \mathbf{Q} = \mathbf{Q}^T 3\mathbf{I} \mathbf{Q} \quad (6)$$

$$\mathbf{Q}^T \mathbf{M} \mathbf{N} \mathbf{Q} + \mathbf{Q}^T \mathbf{N} \mathbf{M} \mathbf{Q} = 3\mathbf{I} \quad (7)$$

$$\Rightarrow \mathbf{Q}^T \mathbf{M} \mathbf{N} \mathbf{Q} = (\mathbf{Q}^T \mathbf{M} \mathbf{Q}) (\mathbf{Q}^T \mathbf{N} \mathbf{Q}) \quad (8)$$

$$\Rightarrow \mathbf{Q}^T \mathbf{N} \mathbf{M} \mathbf{Q} = (\mathbf{Q}^T \mathbf{N} \mathbf{Q}) (\mathbf{Q}^T \mathbf{M} \mathbf{Q}) \quad (9)$$

let

$$\mathbf{Q}^T \mathbf{N} \mathbf{Q} = \mathbf{N}' \quad (10)$$

$$\Rightarrow \mathbf{N} = \mathbf{Q} \mathbf{N}' \mathbf{Q}^T \quad (11)$$

$$\Rightarrow \mathbf{Q}^T \mathbf{M} \mathbf{N} \mathbf{Q} = \mathbf{D} \mathbf{N}' \quad (12)$$

$$\Rightarrow \mathbf{Q}^T \mathbf{N} \mathbf{M} \mathbf{Q} = \mathbf{N}' \mathbf{D} \quad (13)$$

From (11) and (12) (7) can be written as

$$\mathbf{D} \mathbf{N}' + \mathbf{N}' \mathbf{D} = 3\mathbf{I} \quad (14)$$

let $(\mathbf{D} \mathbf{N}')_{i,j}$ be the entry of i th row and j th column of the matrix $\mathbf{D} \mathbf{N}'$. As $\mathbf{D}$ is a diagonal matrix

$$(\mathbf{D} \mathbf{N}')_{i,j} = \mathbf{D}_{i,i} \mathbf{N}'_{i,j} \quad (15)$$

let $(\mathbf{N}' \mathbf{D})_{i,j}$ be the entry of i th row and j th column of the matrix $\mathbf{N}' \mathbf{D}$. As $\mathbf{D}$ is a diagonal matrix

$$(\mathbf{N}' \mathbf{D})_{i,j} = \mathbf{D}_{j,j} \mathbf{N}'_{i,j} \quad (16)$$

$$\Rightarrow (\mathbf{D}_{i,i} + \mathbf{D}_{j,j}) \mathbf{N}'_{i,j} = 3 (\mathbf{I})_{i,j} \quad (17)$$

for $i \neq j$

$$(\mathbf{D}_{i,i} + \mathbf{D}_{j,j}) \mathbf{N}'_{i,j} = 0 \quad (18)$$

$$\Rightarrow \mathbf{N}'_{i,j} = 0 \quad (19)$$

for $i = j$

$$(\mathbf{D}_{i,i} + \mathbf{D}_{j,j}) \mathbf{N}'_{i,j} = 3 \quad (20)$$

$$\Rightarrow \mathbf{N}'_{i,i} = \frac{3}{2\mathbf{D}_{i,i}} \quad (21)$$

From (3)

$$\Rightarrow \text{The diagonal values of } \mathbf{N}' \text{ are } \frac{3}{2} \text{ and } \frac{3}{4} \text{ and } \frac{1}{2} \quad (22)$$

In (11) $\mathbf{Q}$ is an orthogonal matrix and $\mathbf{N}'$ is a diagonal matrix so we can say that (11) is the eigenvalue decomposition of $\mathbf{N}$
 $\Rightarrow \mathbf{N} = \mathbf{Q}\mathbf{N}'\mathbf{Q}^\top$

$$\mathbf{N}' = \text{diag}(\lambda_1, \lambda_2, \lambda_3) \quad (23)$$

$$\Rightarrow \lambda_1 = \frac{3}{2} \text{ and } \lambda_2 = \frac{3}{4} \text{ and } \lambda_3 = \frac{1}{2} \quad (24)$$